

Wilson Schwegler

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Livermore, California

EDUCATION

University of California, Davis, Davis, CA

Master of Science, Computer Science

Expected 3/2027

Occidental College, Los Angeles, CA

Bachelor of Arts, Computer Science and Economics, cum laude

Received 5/2025

SKILLS

Languages: Python, SQL, Java, C++, C, Kotlin, HTML

Libraries: PyTorch, TensorFlow, scikit-learn, YOLO, Pandas, NumPy, D3.js, Pybaseball

WORK EXPERIENCE

Lawrence Livermore National Laboratory, *Graduate Student Intern*, Livermore, CA (6/2025–8/2025, 6/2026–9/2026)

- Developed an in situ monitoring system for nuclear fusion experiments using computer vision to track high-density carbon target capsules during chemical vapor deposition.
- Developed quantitative metrics to characterize capsule movement and interactions, linking experimental behavior to capsule quality and enabling faster screening of manufacturing runs.
- Applied linear, ridge, and lasso regression to small sample experimental data to predict capsule quality from computer vision derived metrics, identifying movement and interaction patterns associated with higher quality and informing new manufacturing conditions.
- Designed a user interface and ML-assisted data labeling tool enabling non-technical users to analyze experiments and efficiently curate training datasets.
- Accelerated processing speeds by up to 2.3× through multi-GPU support, multiprocessing, and code refactoring.
- Presented weekly updates to 5–8 staff scientists and communicated results at a summer research poster session.

Lawrence Livermore National Laboratory, *Computing Scholar Intern*, Livermore, CA (6/2024–8/2024)

- Collaborated with the Quantum Simulations Group on predictive modeling tasks.
- Applied supervised and unsupervised learning methods to analyze the bioactivity in PFAS molecules.
- Utilized neural and graph neural networks to predict the X-ray absorption spectra of amorphous carbon structures.
- Presented research findings at two summer research poster sessions to staff scientists and researchers.

Lawrence Livermore National Laboratory, *Physics with Phones App Developer*, Livermore, CA (6/2023–8/2023)

- Led the UI development for the Android version of the app.
- Created the graphing fragments, enabling users to visualize complex data trends effectively.
- Collaborated with a team of 5 members to refine app features and ensure seamless functionality.

Occidental College Athletics, *Games Management Assistant*, Los Angeles, CA (9/2021–5/2025)

- Supported multiple athletic programs by assisting with scoring, stat recording, and event setup.

HONORS AND AWARDS

- **Graduate Student People's Choice Award**, NIF&PS Scholars Program, LLNL (2026)
- **Department Letter of Commendation**, Economics Department, Occidental College (2025)
- **Pass with Distinction**, Economics Department, Occidental College (2025)
- **Bill Anderson Coaches' Award**, Occidental Baseball (2025)
- **SCIAC All-Academic Team**, Occidental College (2023, 2024, 2025)
- **Dean's List**, Occidental College (2021, 2023, 2024, 2025)

ACTIVITIES

Baseball, Occidental College (*Team Captain*)

- Led and motivated teammates, fostered teamwork, and served as a liaison between players and coaches.

RELEVANT COURSEWORK

Computer Science: Machine Learning, Information Visualization, Artificial Intelligence, Scientific Computing, Visual Analytics, Data Structures, Algorithms Analysis, Databases, Computer Graphics, Justice and Equity in Technology

Economics: Game Theory, Econometrics, Macroeconomic Policy and Financial Markets, Microeconomic Theory, Macroeconomics, Empirical Financial Economics, and Environmental Economic Policy